

2.3 Logic as a Knowledge Representation Framework

Logic is the science of true reasoning and is an important concept in computer science.

Why we use Logic?

- It helps computers store, understand, and use information correctly.
- Using logic, complex information can be represented in a simple way.

TIDBIT Logic gives computers a way to “think” about facts and make decisions, almost like a brain.

Logic in Computer Science

In computer science, logic means using rules to make correct decisions.

- Logic helps computers understand true and false conditions.
- It allows systems to process information correctly.
- Logical statements are used in programs and algorithms.
- These statements help control the flow of programs.
- Logic is the basis of many computer operations.
- It helps computers think in a structured way.

Role of Logic in Reasoning and Decision Making

Logic plays an important role in reasoning and decision making.

- It helps computers and humans **draw correct conclusions**.
- Decisions are made based on facts and rules.
- It removes confusion and guesswork.
- Logical reasoning helps solve problems step by step.
- Computers use logic to choose the correct action.
- Logic ensures that decisions are consistent and accurate.

Where we use it?

- It is used in expert systems and artificial intelligence.

CLASS ACTIVITY Discuss how logic helps in making decisions. Students make simple “if-then” statements about school life. Example: “If it rains, then we will stay indoors.”

Short Questions

1. What is logic?
2. Why is logic important in computer science?
3. How does logic help computers?

4. What is the role of logic in reasoning and decision making?
5. How does logic remove confusion in decision making?
6. Where is logic used in computer systems?
7. Give one example of how logic helps in daily life decisions.
8. What does logic allow systems to process correctly?